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The Value of Apology: How do Corporate Apologies Moderate the Stock Market Reaction to Non-Financial Corporate Crises?

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Abstract

In a crisis, managers are confronted with a dilemma between their ethical responsibility to respond to victims and their fiduciary responsibility to protect shareholder value. In this study, we use a unique and comprehensive dataset of 223 non-financial crises between 1983 and 2013 to investigate how corporate apologies affect stock prices. Our empirical evidence shows that the stock price response from apologizing depends on the firm's level of responsibility for the crisis. We find that to protect shareholder value, management needs to match its formal response strategy with the degree of its responsibility for the crisis. Further, failing to match the proper response is harmful to shareholders: Both apologizing when the firm is not directly responsible for the crisis and failing to apologize when the firm is directly responsible reduce shareholder wealth. However, apologizing for a crisis when the firm is directly responsible mitigates losses to shareholder value that arise because of the crisis, as does refraining from apologizing when the firm is not directly responsible.

Keywords Apology · Crisis management · Event study · Expectancy violation theory · Investor's behavior · Situational crisis communication theory

Introduction

In 1987, Chrysler was indicted for tampering with odometers. Executives were using vehicles with disabled odometers for thousands of miles of personal travel before connecting the odometer and selling the vehicle “as new.” Senior administrators apologized, and the market's initial response was a two percent decline in cumulative abnormal return (CAR), but after 20 days, the CAR increased 9%.¹ Conversely, in 2011 Blackberry users suffered a 4-day service outage due to a hardware failure of an independent satellite. Senior administration apologized, the initial CAR fell 3.3%, and after 20 days, it had dropped 29%. These are examples of two crises,

two apologies, but dramatically different impacts on price. Our research provides answers to the disparity observed in these cases by investigating the financial consequences surrounding non-financial organizational crises. The key elements we consider are the firm's degree of responsibility for the crisis and the organization's response to the crisis. In particular, we are interested in how corporate apologies affect the stock price reaction to a crisis: Does a corporate apology mitigate or exacerbate the typically negative response to an organizational crisis, conditional on the firm's degree of responsibility for the crisis?

Due to the adverse consequences and unexpected nature of a crisis, prior research finds that the stock market tends to react negatively to such events. For example, Broder and Morrall (1991) observe a 2.57% abnormal decrease in share price in the first ten days after a fatal accident, and Ho et al. (2013) find that airplane crashes lead to abnormal share price losses of 1.85% in the first 10 days. While prior research demonstrates an adverse market reaction to crises, there has been little research on how investors respond to

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¹ We use CAR and not raw returns to mitigate the influence of other trends such as decreasing market share due to competitors or contemporaneous events such as earnings announcements. We provide a more detailed discussion of CAR in the “Methods and Data” section.

communication from the organization about the crisis. Marcus and Goodman (1991) categorize responses to crises as being either defensive (e.g., denial) or accommodative (e.g., apology) and find that for corporate scandals, investors react more positively to accommodative communication, but for accidents, investors react more negatively to accommodative communication. One possible confounding factor in Marcus and Goodman's study is that investor reaction to corporate communication about a corporate scandal cannot always disentangle the investor from the victim, because in some cases they are not mutually exclusive. For example, in a plane crash the primary victims are those on the plane and their immediate families; investors may be hurt financially, but they are not the direct victims of the crisis. However, in cases of financial misdeeds, such as fraud or bribery, shareholders are direct victims (possibly in conjunction with other stakeholders such as an alternative supplier who could have otherwise obtained a contract or a lender who issued a loan based on false information). Similarly, Chance et al. (2015) find that firms that make announcements accepting blame for poor performance subsequently outperform those firms that try to attribute blame to external sources. These studies provide compelling insight into investor reaction to a crisis; however, they do not exclusively consider situations in which the crisis does not directly victimize shareholders. The distinction is important, since apologizing to victims creates an emotional response, so the stock price response when shareholders are victims may correspond to a behavioral as opposed to a rational reaction (Jin et al. 2012).

While there is limited archival empirical research looking at organizational communication in response to a crisis, there is a large body of literature based on experimental psychology methods that explores how individuals react to crises and crisis responses. Many of these studies rely on the Situational Crisis Communication Theory (SCCT) from the crisis management literature (Coombs 2007; Ma and Zhan 2016). This theory postulates that an organization's stakeholders respond more negatively as their perception of the organization's responsibility for a crisis increases (Jorgensen 1996; Coombs and Holladay 1996, 2002, 2008; Claeys et al. 2010). The theory further postulates that the negative response can be attenuated by the organization appropriately accepting responsibility. In particular, SCCT advocates communicating an acceptance of responsibility equal to the degree of the firm's responsibility. Failure to communicate an appropriately matched response to the crisis could exacerbate the adverse stakeholder reaction to the crisis, consistent with the Expectancy Violations Theory (EVT) (Burgoon 1993). Following this literature, we classify crises as preventable when the firm has a high degree of direct responsibility for the crisis, and non-preventable when the cause of the crisis is due, in large part, to external factors. In our introductory examples, tampering with odometers would

be classified as a preventable crisis, whereas a service outage due to an external hardware failure would be non-preventable. While the theory is easily applied to quantitative analysis (Sellnow and Seeger 2013), to the best of our knowledge, ours is the first study to take SCCT out of the laboratory and test it using real data in a context outside of the consumer response literature. We investigate how the SCCT applies to investor response for non-financial crises using the archival methods of empirical finance.

Our study uses a sample of 223 unique organizational crises covering the period 1983–2013. Of these 223 crises, the firms involved issue apology statements in 46 cases, of which 38 apologize within 20 days of the crisis. We construct our sample by using crises studied in previous research, supplemented with hand-collected data from other sources. We use an event study method as well as cross-sectional regression to test our hypotheses around the crisis event. We perform robustness tests by controlling for endogeneity using a two-stage least squares approach and by running a second event study around the apology date using only firms that apologized for the crisis. After controlling for factors related to the history of previous crises, the company's reputation, and the abnormal publicity that arises from the crisis (among others), we find a significant negative relation between share price and preventable crises. Similarly, we find a significant negative association between share price and the occurrence of fatalities during the crisis. We also show that matching the firm's response to its degree of responsibility is necessary for diminishing the negative stock price reaction to the crisis. In particular, we report a significant negative relation between CAR and mismatched responses such as apologizing for a non-preventable crisis or failing to apologize for a preventable crisis, and we show a positive relation for matched responses such as apologizing for a preventable crisis or refraining from apologizing for a non-preventable crisis. While we are confident that we have compiled a comprehensive sample and that our analysis is rigorous and thorough, we note that corporate apologies are becoming more common over time, so future studies may yield varying results.

By taking theory out of the lab and testing its external validity with data in the field, we make several important contributions to the literature. We substantiate prior theory about SCCT by showing that organizational responsibility is inversely related to share price. As an organization's responsibility for a crisis increases, its stock price reaction to the crisis is more negative (Ma and Zhan 2016). We also contribute to the literature by using archival data to quantify the effect of an apology on share prices (Koehn and Goranova 2018; ten Brinke and Adams 2015). We show that apologizing is beneficial to shareholders if and only if the firm is directly responsible for the crisis. This offers new insight into the importance of adhering to the matching

hypothesis put forward by the SCCT (Coombs 2007; Ma and Zhan 2016).

We organize the remainder of our paper as follows: In “Theoretical Framework and Hypotheses”, we review prior research on crises and apologies before introducing the theoretical framework used to derive our hypotheses. “Methods and Data” provides a description of our sample and explains the methods we use for empirical analysis. “Results” presents our results and offers some additional analysis. “Conclusion” includes conclusions, limitations, and recommendations for future research.

Theoretical Framework and Hypotheses

Situational Crisis Communication Theory

Corporate crises are significant economic events that can have negative consequences for victims, shareholders, and other stakeholders (Coombs 2007). To understand why shareholder response varies systematically among crises, we rely on the Situational Crisis Communication Theory (SCCT) as a conceptual framework to derive our hypotheses. It is especially useful because it offers a typology of crises and best responses that is prescriptive in nature.

SCCT is built on the principles of attribution theory. Attribution theory suggests that individuals have a desire to assign blame or responsibility to someone or something for events that they witness.² Weiner’s (1985) attribution theory model states that people make attributions along three dimensions: stability, the locus of control, and controllability. Stability refers to how the event relates to past events. Locus of control involves the degree to which internal or external forces cause the development, and controllability refers to whether the event was intentional or not. These three factors interact as individuals make causal inferences about an event to determine the extent of the other party’s responsibility.

Crisis type, Organizational Responsibility, and Share Price

Originally, SCCT categorized crises into four categories: faux pas, terrorism, accidents, and transgressions (Coombs 1995). This list was refined over time to arrive at the current version consisting of ten types of crises grouped into three clusters that are based upon similar attributes of crisis responsibility (Coombs and Holladay 2002; Coombs 2007). The first group is the victim cluster, in which the firm clearly has little responsibility (the cause is external and unintentional). The second group is the accidental cluster, for crises

in which the firm is neither apparently responsible nor unambiguously a victim (the cause is internal or unclear and unintentional). The third group is the preventable cluster, which accounts for crises where the responsibility clearly resides with the company or its agent(s) (the cause is internal and intentional). See the first column of Table 1 for a categorized list of crisis type clusters and brief definitions.

Research looking at crises shows that regardless of the type of crisis, a firm’s share price tends to be negatively affected when it suffers a crisis.³ For example, Walker et al. (2005) find that the airlines associated with an aviation disaster suffer cumulative abnormal returns of -3.18% in the first 14 days after a disaster.⁴ Bonini and Boraschi (2011) show that corporate scandals lead to large losses in the first day of the crisis (-4.00%) which continually worsen for up to 10 days after the scandal announcement (-6.20%). Other studies focusing on a cross section of crisis types also find significant losses associated with firms in crisis (Chance et al. 2015; Knight and Pretty 1999; Marcus and Goodman 1991).

Research that focuses more specifically on the relation between organizational responsibility for a crisis and stakeholder behavior reveals a negative association between responsibility and stakeholder actions. Claeys et al. (2010) show that reputation is lower for preventable crises, but there is not a significant effect on reputation for accidental and victim crises. Jorgensen (1996) concludes that consumer predilection to punish an organization is positively related to the firm’s responsibility for the crisis. Coombs and Holladay (2007) find that higher levels of responsibility for a crisis lead to lower purchase intentions and a greater likelihood of speaking unfavorably about the organization. Similarly, in a meta-analysis of experimental studies using SCCT, Ma and Zhan (2016) find that organizational responsibility is negatively related to organizational reputation. Finally, recent research using public companies in China finds a negative relation between crisis responsibility and firm value in the wake of a crisis (Wei et al. 2017). We likewise expect an

² See Fiske and Taylor (1991) for an overview and several variations of attribution theories.

³ A crisis can affect firm value by directly reducing cash flows (crisis disrupts operations) or indirectly by increasing the riskiness of the cash flows (increased lawsuits or revealing an increased risk for a specific firm or industry), or both. We do not attempt to distinguish how a crisis or an apology affects these components independently, but rather we are concerned more generally with the overall effect on share price.

⁴ Walker et al. (2005) also consider long-term returns and find that by 12 months, the abnormal returns are -1.73% and by 24 months, they are 5.74%, (although in both cases they are not significantly different from 0). One could argue that this long-term reversal is due to an overreaction and subsequent correction, but the fact that it takes over a year to correct does not lend credibility to such a claim. In our analysis, we include multiple event windows between 2 and 20 days and rule out over- or under-reaction and subsequent correction as the cause of our results.

Table 1 Crisis type with corresponding crisis response strategy

Crisis types	Crisis response strategies
Victim cluster Natural disasters (acts of nature damage an organization) Rumors (false and damaging information about an organization is being circulated) Workplace violence (current or former employee attacks current employees onsite) Malevolence/product tampering (external agent causes damage to an organization)	Denial posture Attacking the accuser (crisis manager confronts the person or group claiming something is wrong with the organization) Denial (crisis manager asserts that there is no crisis) Scapegoating (crisis manager blames some person or group outside the organization for the crisis)
Accidental crisis cluster Challenges (stakeholders claim an organization is operating in an inappropriate manner) Technical-error accidents (a technology or equipment failure causes an accident) Technical-error product harm (a technology or equipment failure causes a product to be harmful to users)	Diminishment posture Excusing (crisis manager minimizes organizational responsibility by denying intent to do harm and/or claiming inability to control the events that triggered the crisis) Justification (crisis manager minimizes perceived damage caused by the crisis)
Preventable crisis cluster Human-error accidents (human error causes an accident) Human-error product harm (human error causes a product to be harmful to users) Organizational misdeeds (stakeholders are deceived and/or laws or regulations are violated by management)	Rebuilding posture Compensation (crisis manager offers money or other gifts to victims) Apology (crisis manager indicates the organization takes full responsibility for the crisis and asks stakeholders for forgiveness)

adverse response from shareholders during a crisis that is stronger for crises where the firm's responsibility is higher. That is, we propose our first hypothesis as a calibration with prior research that contributes to the empirical relationship evidence between crisis responsibility and share price.

Hypothesis 1 The share price response to a crisis is negatively related to the firm's responsibility for the crisis.

Organizational Apology

The prescriptive nature of SCCT leads to recommendations about how an organization should respond to a crisis based on its level of responsibility. The SCCT recommends seven crisis response strategies grouped into three types: deny, diminish, and rebuild. The crisis response matches the crisis type according to the level of responsibility: victim implies deny, accident implies diminish, and preventable implies rebuild. Column 2 of Table 1 offers a brief description of the crisis responses within the three clusters.

How firms respond to a crisis has been shown to influence the market's behavior. For example, Marcus and Goodman (1991) look at how the use of an accommodative versus defensive response strategy affects returns while Lee et al. (2004) and Chance et al. (2015) focus on how internalizing blame affects share price. The results suggest that investors react more positively to responsibility accepting

communication when the firm is perceived to be responsible for the crisis.

The most common way a firm can accept responsibility for a crisis is by offering an apology. An apology is a social device that is given for events that transpired in the past, but the decision to apologize is primarily driven by future considerations (Wohl et al. 2011; Ho 2012). Motivations to apologize include a desire to restore trust and reputation, mitigate negative feelings towards the transgressor, decrease punishment, and to re-engage with the victim. A company may also apologize to try to retain customers and other stakeholders that were harmed by the crisis. Patel and Reinsch (2003) find that an apology cannot shield against actual damages levied by a court, but it can be used by the defense to avoid punitive damages. Similarly, Robbennolt (2003) finds that apologizing can give increased bargaining power to the apologizer. Numerous studies find that apologies can mitigate the victim's negative emotions towards the harming individual (Ohbuchi et al. 1989; Jorgensen 1996; McDonald et al. 2010; Lee and Chung 2012) and some research finds that apologies improve the apologizer's reputation (Claeys et al. 2010; Rasso 2014). The effectiveness of an apology in mitigating stock losses is positively related to the content of the apology but negatively related to the emotion conveyed in the apology (Koehn and Goranova 2018).

Apologizing can also have costs or adverse side effects. For example, the public nature of apologizing draws

attention to a problem that initially may have had a limited audience (Friedman 2006), and increased exposure can bring increased scrutiny. Apologizing can also have an adverse effect in situations where the severity of harm is very high by increasing negative views of the apologizer (Ohbuchi et al. 1989). It can also lead to increased levels of blame being assigned to the apologizer (Jorgensen 1996; Coombs and Holladay 2002; Zechmeister et al. 2004), although counter evidence also exists (Cornell et al. 2009; McDonald et al. 2010).

The possibility of both positive and negative consequences for apologizing leads to conflicting advice offered by public relations departments, crisis management consultants, and corporate lawyers. The public relations view favors open discourse with stakeholders, offering as much information as possible, including an apology when warranted, whereas the legal view favors limited disclosure thereby offering little information that could potentially be used in a lawsuit (Kaufmann et al. 1994; Fitzpatrick and Rubin 2001; Patel and Reinsch 2003; Robbennolt 2003; Coombs 2007; Coombs and Holladay 2007).

Matching Crisis Response Strategies to Crisis Type

Some research has shown that individuals expect and are most satisfied with an apology relative to other crisis response strategies (Brocato et al. 2012), leading some to prescribe apology as the go-to response for a crisis (Benoit and Drew 1997; Schweitzer et al. 2015). However, Coombs (2007) argues that an apology should only be offered when the firm has a high level of responsibility for the crisis. These competing recommendations lead to confusing advice about when an apology should and should not be given by a company.⁵

According to SCCT, firms must match their crisis response to the level of their perceived responsibility for the crisis. However, the efficacy of matching the response to the crisis type has mixed empirical support (Ma and Zhan 2016). Coombs and Holladay (1996) find that matched responses lead to a higher reputation than mismatched responses, whereas Claeys et al. (2010) conclude that matching the response to the crisis type has no consequence to the organization's reputation. Similarly, Bundy and Pfarrer (2015)

argue that a firm's social approval, a concept that captures the general affinity, attractiveness, or likeability of a firm, is lower when the firm's crisis response does not match the level of responsibility they bear for the crisis. This negative consequence from mismatching applies both to situations where the firm accepts little or substantial responsibility through their choice of crisis response. However, we contend that mismatching crisis response to crisis type can lead to harmful effects as investors' expectations are violated.

Expectancy Violations Theory

Expectancy Violation Theory (EVT) (Burgoon 1978, 1993) offers an alternate explanation for the consequences of mismatching crisis response to crisis type. EVT, a theory about communication and societal norms, suggests that individuals form expectations about other people's behavior (Burgoon 1993). Recent research has used EVT to explain such things as how investors' judgments and decisions are affected by the alignment or misalignment of a firm's accounting choices with investors' expectations (Clor-Proell 2009) and anticipatory impression management in the context of company acquisitions (Graffin et al. 2016).

The purpose of EVT is to explain individuals' behavior when their expectations are not met, in which case they engage in interpretation of the violator's actions, often leading the person to behave in a negative way (Burgoon 1993). Kim (2014) applied EVT in an experiment to gauge how people respond to expectancy violations in the context of the BP oil spill crisis. The author finds that violations of expectations lead to higher assessments of uncertainty about BP's future performance, higher levels of responsibility for the crisis, and a lower reputation. The author also concludes that the apology offered by BP positively influenced BP's reputation and was better than providing an excuse or no message at all.⁶ Thus, we predict that shareholders will react more negatively when the crisis response does not match the crisis type but will react less negatively when the crisis response does match the crisis type.

Hypothesis 2a Apologizing exacerbates the negative share price response for non-preventable (victim and accidental) crises.

Hypothesis 2b Apologizing mitigates the negative share price response for preventable crises.

⁵ Intuitively, apologizing for a crisis yields costs and benefits. Costs are associated with taking responsibility and correspond to legal obligations (or an implicit promise of compensation). Benefits come from improved reputation and stakeholder relations. Indirect costs and benefits can arise from how the apology signals the future: Acknowledgment of fault suggests past incompetence, which could signal future incompetence. On the other hand, apologizing could signal understanding of a past mistake, reducing the likelihood of repeating it. SCCT suggests that matching response to responsibility helps ensure that the benefits outweigh the costs.

⁶ Kim's (2014) results have limited generalizability due to the single high profile case used in the experiment and by having participants recall information that may have been distorted due to "the curse of knowledge" (Camerer et al. 1989).

Methods and Data

Sample Construction

Our paper uses empirical evidence to study stock market reaction to organizational crises and subsequent apologies. We identified 341 different crises for the period 1983 to 2013 using a thorough and exhaustive search of a variety of sources. The crises were determined from the following sources and in the following order: (1) prior peer-reviewed articles (Chance and Ferris 1987; Borenstein and Zimmerman 1988; Mitchell and Maloney 1989; Marcus and Goodman 1991; Long and Rao 1995; Knight and Pretty 1999; Capelle-Blancard and Laguna 2010; Carpentier and Suet 2015; Krieger and Chen 2015), (2) Institute for Crisis Management annual crisis report for the years 1993–2012, (3) government safety bodies (National Transportation Safety Board & U.S. Chemical Safety and Hazard Investigation Board), and (4) private correspondence with a respected academic in the crisis management and public relations area. Other sources and methods were also investigated such as internet searches for terms “corporate crisis” or “corporate apology” as well as attempting to draw on the power of social networks. These alternate connections did not yield any new observations.

The goal is to create a comprehensive sample of different types of crises across various industries and time periods, but we exclude 65 accounting and securities fraud events for two reasons. First, in cases of fraud, the primary victims are the investors themselves in contrast to the other crises in our sample where non-investor stakeholders, such as customers, employees, communities, or the environment are the primary victims.⁷ Thus, connection to the crisis and its direct impact on investors may lead to different reactions compared to a crisis where there is a clearly delineated victim group and investor group. Second, we were unable to identify any cases where an organization offered an apology for having committed fraud; thus, the inclusion of accounting and securities fraud would provide no testable effect of an apology for this type of crisis.

We collect reputation data from Fortune’s “World’s Most Admired Companies.” Stock return data and accounting information are drawn from the Center for Research in Security Prices (CRSP) and COMPUSTAT, respectively. Missing reputation data, at both the firm and industry level, led to the exclusion of 30 observations, and an additional 17

observations were dropped due to missing CRSP or COMPUSTAT information. Finally, to avoid confounding effects, we eliminate six observations that occurred within 6 months of a previous unrelated crisis, yielding a final sample of 223 different crises. While most are US firms, 23 firms are headquartered outside the US.⁸

Variables

We investigate how the stock market reacts to the news of a crisis and how this response is affected by whether or not the firm offers an apology. We also examine how the market reacts to an apology provided by the company in response to the crisis. To test our hypotheses about how crises and apologies affect a firm’s share price we calculate cumulative abnormal returns (CAR) following standard event study methods as outlined by Brown and Warner (1980, 1985). In addition, we run a cross-sectional regression to control for crisis and firm characteristics that likely affect the financial response to the crisis and apology. We provide variable definitions in Table 2.

Dependent Variable

Our dependent variable is the CAR calculated for each firm for the specified event window, $CAR_i(\tau_0, \tau_1)$. The event window is defined as the number of trading days from the crisis date (τ_0) to a pre-specified endpoint (τ_1). In our study, τ_1 ranges from day 2 to day 20 leading to five event windows: [0, 2], [0, 5], [0, 10], [0, 15], and [0, 20] using the crisis as the event date 0. The estimation window used to calibrate “normal returns” consists of the 255 trading days between -275 (τ_{-2}) and -20 (τ_{-1}) prior to the crisis.

The estimation window is used to calculate the expected return under normal market conditions (no news), as measured by the single-index market model. We use the value-weighted index from the CRSP database as a proxy for the market portfolio and estimate the intercept and market beta by regressing daily firm returns on daily market returns over the estimation window.⁹ We then use the estimated intercept and beta to generate expected returns during the event window. The abnormal returns (ARs) are the difference between actual daily returns and expected returns as predicted from the single-index model during the event window. To the

⁷ Because of financial loss, investors could be considered as at least secondary victims of all crises. However, there is an important distinction, since, as the primary victims, they would be the wronged party who is owed the apology, whereas as secondary victims, an apology to shareholders would be inappropriate.

⁸ For robustness, we run all of our analysis excluding the 23 non-US firms. Dropping these 23 firms does not alter our results.

⁹ For robustness, we use several other models to calculate the expected return. These include the equally weighted CRSP index to proxy the market portfolio in the single-index model, the Fama–French three-factor model, and the Carhart four-factor model. We also use a GARCH(1,1) model to account for potential heteroskedasticity. We find marginal differences in event window CARs, but no

Table 2 Variable definitions

Variable name	Variable definition [source]	Predicted sign
Dependent variable		
CAR	Cumulative abnormal return is calculated for the event window using event study methodology as outlined by Brown and Warner (1980, 1985)	
Independent variables		
Preventable	= 1 if the crisis is classified as one of the three crisis types that make up the preventable crisis cluster according to SCCT	–
Apology	= 1 if the firm apologizes on any day within the event window	–
Apology * preventable	The interaction of the two dummy variables <i>APOLOGY</i> and <i>PREVENTABLE</i>	+
Control variables—crisis level		
Fatality	= 1 if deaths occurred because of the crisis	–
History	= 1 if the firm has had previous crises that are captured in the sample	–
Reputation	Most recent reported score in Fortune magazine's annual "World's Most Admired Companies" ranking; for firms missing a raw score the industry average is used. [FORTUNE]	+
Publicity	The log of average daily media coverage during the event window estimated with a keyword search in abstracts of major newspapers minus the log of average daily media coverage during the estimation window. [ABI GLOBAL]	–
Volume	Abnormal trading volume is calculated for the event window using event study methodology similar to how CAR is calculated. [CRSP]	–
Control variables—firm level		
Size	The log of market capitalization the day prior to the crisis. [CRSP]	+
Leverage	Total liabilities divided by total assets from the most recent financial statements. [COMPUS-TAT]	–
Market-to-book	Market value of equity the day before the crisis divided by book value of common equity from the most recent financial statements. [COMPUTSTAT]	?
IND	Category variable that captures the 12 industries in the Fama–French industry classification schema	
Year	Category variable that captures three-year increments between 1983 and 2013, resulting in ten categories	

extent that the stock price may take some time to reflect information about the crisis, we use the cumulative abnormal return (CAR), which is the summation of ARs across the event window.

Independent Variables

We have two key variables, *PREVENTABLE* and *APOLOGY*, that, by themselves and in interaction with one another, allow us to test our hypotheses. *PREVENTABLE* is a dummy variable indicating that the crisis is preventable because the firm bears a high degree of responsibility for the crisis. We categorize crises as preventable or non-preventable using one of the two methods. If the crisis has an associated government report that identifies who is at fault, we use the government's description to categorize the crisis as either preventable (firm is at fault) or non-preventable (firm

is not at fault). For example, the National Transportation Safety Board investigates airline and railroad crashes and their reports identify who (e.g., pilot-error) or what (e.g., engine failure) was the cause of the crash. In our sample, 99 crises have a report from a government entity. Second, for cases where no investigative body produced a publicly available report, we searched news articles that described the crisis to determine the cause of the crisis and assign it to one of the crisis categories listed in Table 1.¹⁰ Hypothesis 1 states that firms with a preventable crisis should suffer larger losses compared to companies with a non-preventable crisis.

APOLOGY is a dummy variable indicating that the company has apologized at some time during the event window. For example, if a firm apologizes on day 3, then *APOLOGY* will be 0 for event window [0, 2], but 1 for event windows [0, 5] and beyond. To identify and locate a firm apology,

Footnote 9 (continued)

significant differences between the various event study outcomes that affect our interpretation of the results.

¹⁰ To avoid subjective measurement error in our classification of crises that did not have an associated government report, we repeated all our analyses (unreported) with only those crises that have an official government report. This subsample yields similar, albeit slightly weaker results.

we follow Chance et al. (2015) and search for news articles around the crisis date that contains the firms name, reference to the crisis, and one of the following keywords: “apology,” “apologize,” or “sorry.” Hypotheses 2a and 2b state that firms that apologize for a non-preventable crisis will suffer larger losses, but firms that apologize for a preventable crisis, measured by the interaction between apology and preventable crisis, will have reduced losses.

Control Variables

In addition to the key test variables, we also include control variables related to the crisis and the firm. FATALITY is a dummy variable indicating that the crisis is responsible for at least one death.¹¹ To link a crisis with fatalities, we use the same procedure that we used to categorize crisis as preventable or non-preventable. First, we look for official government reports that describe the crisis and, in their absence, we search news articles for any discussion of fatalities or similar words (e.g., “deaths”). We expect that crises with fatalities should lead to an increase in crisis severity, which will result in larger losses for the associated firm.

We control for a firm’s prior history of crises with a dummy variable (HISTORY) indicating that the firm has had at least one other crisis observation within our sample in the last five years.¹² We expect a negative association with CAR since investors should frame the current crisis as more negative and the firm more responsible if it has a history of previous crises (Coombs 2007). We control for the reputation of the firm (REPUTATION) by using the most recent reported score in Fortune magazine’s annual “World’s Most Admired Companies” ranking. There are 68 unranked firms and we use the appropriate industry average to proxy for the missing firm reputation.¹³ Aaker et al. (2004) find that company brand reputation affects how consumers respond to corporate transgressions. We expect that companies with higher relative scores, hence better reputations, can better deflect negative behavior, which should yield a positive relation with CAR (Fombrun and Shanley 1990). We control for the abnormal level of publicity arising from the crisis (PUBLICITY) with the log of average daily media coverage during the event window estimated using a keyword search

in abstracts of major newspapers minus the log of average daily media coverage during the estimation window (Fombrun and Shanley 1990; Knight and Pretty 1999). We use ABI/INFORM Global to search major news sources including the Wall Street Journal and the New York Times. We expect higher levels of media coverage to make the negative aspects of the crisis more salient to investors, resulting in a decrease in share price. We control for abnormal trading volume (VOLUME) by using the same event study methodology to identify cumulative abnormal trading volume for each event window. Bamber et al. (2011) notes that the coefficient from regressing trading volume on price change captures differential precision of pre-disclosure information (information asymmetry). Although we regress abnormal return on abnormal trading volume, we expect the coefficient on abnormal trading volume to capture varying interpretations about the uncertainty and risk of the financial consequences of the crisis, and hence to be negatively related to CAR. Furthermore, we control for several firm characteristics prevalent in prior research, including firm size (SIZE), which is measured by the log of market capitalization on the day prior to the crisis (Walker et al. 2005),¹⁴ leverage (LEVERAGE), total liabilities divided by total assets from the most recent financial statements prior to the crisis, market to book (MARKET-TO-BOOK), the market value of equity the day before the crisis divided by book value of common equity from the most recent financial statements prior to the crisis (Fombrun and Shanley 1990), as well as both industry fixed effects based on Fama and French’s (1997) 12 industries,¹⁵ and year fixed effects, categorized into 10 groups as shown in Panel C of Table 3.

Event Study Tests

We use multiple test statistics to examine the CARs for the full sample and the individual portfolios in both the crisis event study and the apology event study. All of our tests are performed using the EVENTUS software available on the Wharton Research Data Services platform. We report parametric (CSECT) and nonparametric (SIGN) tests.¹⁶ The parametric test, CSECT, is a *t* test that adjusts for possible variance increases during the event window. Brown and Warner (1985) note that event studies using daily data tend

¹¹ For robustness, we also measured fatalities as an ordinal variable and our results are unchanged.

¹² The average time for a firm with repeat crises is 4.5 years. For robustness, we reran our analyses using a HISTORY variable that equalled 1 if the firm had a crisis within the previous 1, 2, or 10 years. All three variations of measuring HISTORY had no effect on the results.

¹³ For robustness, we replaced industry average by 0 for missing values, similar to Cao et al. (2012). This change had no effect on our main conclusions about preventable crises and apologies, but it did increase the statistical significance of the REPUTATION coefficients.

¹⁴ For robustness we used the log of total assets as an alternative measure of size. Our results are not sensitive to how firm size is measured.

¹⁵ For robustness we also used the 2-digit SIC code. We find similar results using this alternative industry measure.

¹⁶ Although we report only two test results, we ran two more parametric tests, CDA and BMP, as well as one other non-parametric test, RANK, as part of our analysis. These additional measures did not change our inferences, and hence, for brevity, we report only the CSECT and SIGN test results.

Table 3 Sample characteristics

	Total sample	Apology	Non-apology
Panel A: crisis characteristics			
Victim crisis	13	3	10
Accidental crisis	162	21	141
Preventable crisis	48	14	34
Total	223	38	185
Fatalities	101	5	96
Crisis history	87	14	73
Panel B: Fama–French Industries			
Consumer non-durables	8	4	4
Consumer durables	7	3	4
Manufacturing	12	0	12
Energy	33	5	28
Chemicals	18	1	17
Business equipment	9	4	5
Telecommunications	2	1	1
Utilities	19	0	19
Wholesale, retail and some services	13	7	6
Healthcare, medical equipment, and drugs	18	2	16
Finance, insurance, and real estate	6	1	5
Other	78	10	68
Panel C: year of crisis			
1983–1985	11	0	11
1986–1988	14	3	11
1989–1991	16	2	14
1992–1994	20	1	19
1995–1997	29	4	25
1998–2000	24	2	22
2001–2003	20	3	17
2004–2006	24	3	21
2007–2009	28	6	22
2010–2013	37	14	23

to be non-normal in their distributions due to fat tails and a right skew, which can lead to false rejection rates when using parametric tests. In this case, non-parametric tests are more powerful since they make no assumptions about the underlying distribution. Hence we also use the SIGN test as specified by Cowan (1992).

Cross-sectional Regression

We use cross-sectional regression analysis to investigate the relation between CAR and our main variables while controlling for crisis and firm characteristics. The regression is estimated using ordinary least squares (OLS) with robust Huber–White standard errors:

$$\begin{aligned} \text{CAR}_i(\tau, \tau_{+1}) = & \beta_0 + \beta_1 \text{PREVENTABLE}_i + \beta_2 \text{APOLOGY}_i \\ & + \beta_3 \text{APOLOGY} * \text{PREVENTABLE}_i \\ & + \text{Controls}_i + \varepsilon_i \end{aligned} \quad (1)$$

where control variables include FATALITY, HISTORY, REPUTATION, PUBLICITY, VOLUME, SIZE, LEVERAGE, MARKET-TO-BOOK, industry fixed effects, and year-group fixed effects. Hypothesis 1 predicts that β_1 should be negative, Hypothesis 2a predicts that β_2 should be negative, and Hypothesis 2b predicts that β_3 should be positive.

Results

Sample Frequency and Descriptive Statistics

Table 3 presents the frequencies for characteristics related to crisis type (Panel A), industry (Panel B), and year (Panel C). The sample consists of 22% preventable and 78% non-preventable crises. Approximately 29% of the preventable crises are accompanied by an apology, whereas only 14% of the firms involved in non-preventable crises apologize.

Finding that apologies are more frequent among preventable crises provides some support of the application of SCCT in practice. However, even when the firm is mainly responsible, over 70% of companies choose not to apologize. The sample also includes 101 crises where at least one person died, although only 5% of firms with fatalities apologize. There are 87 instances of repeat crises for the same company, of which 16% offer an apology. Using the twelve Fama–French industrial categories (Panel B), we find that in our sample, apologies occur in all industries except manufacturing and utilities. Apologies are most prevalent for consumer goods, business equipment, and wholesale/retail industries. When sorted by year (Panel C), we find that no firms offered an apology for any of the 11 crises during 1983–1985, but there are apologies in every period after that. Apologies are most prevalent during the period 2010–2013, where the apology rate is approximately 38% compared to only 14% for the prior period 1986–2009, reflecting an increasing trend of corporate apologies.

We present descriptive statistics in Table 4. The mean and median values for our dependent variable CAR are

similar to those in previous research. The CARs continue to decline as the window size increases up until the 15-day window. APOLOGY increases as the window size increases. The abnormal publicity (PUBLICITY) increases up until the 10-day event window, whereas the abnormal volume (VOLUME) increases throughout.

Table 5 provides correlations. To save space, for variables measured at multiple event windows (CAR, APOLOGY, PUBLICITY, VOLUME) we report only the largest correlation among all event windows. Consistent with Hypotheses 1, CAR is negatively correlated with PREVENTABLE (−0.14). We also find that without consideration for the type of crisis, APOLOGY is negatively correlated with CAR (−0.26). The univariate correlations among other variables are consistent with expectations. For example, CAR is negatively correlated with FATALITY, PUBLICITY, and VOLUME, indicating that more severe crises and higher levels of crisis awareness by the public lead to stronger negative reactions by the market.

Table 4 Descriptive statistics

	Mean	Median	Std. Dev.	Min	Max
CAR [0, +2]	−1.70%	−0.97%	6.12%	−47.85%	20.94%
CAR [0, +5]	−2.66%	−1.24%	8.09%	−48.47%	16.10%
CAR [0, +10]	−2.83%	−1.78%	9.18%	−38.95%	26.93%
CAR [0, +15]	−3.41%	−2.39%	10.56%	−44.24%	31.97%
CAR [0, +20]	−2.89%	−2.22%	11.88%	−47.77%	39.10%
APOLOGY [0, +2]	0.072	0.00	0.259	0.00	1.00
APOLOGY [0, +5]	0.108	0.00	0.311	0.00	1.00
APOLOGY [0, +10]	0.143	0.00	0.351	0.00	1.00
APOLOGY [0, +15]	0.161	0.00	0.369	0.00	1.00
APOLOGY [0, +20]	0.170	0.00	0.377	0.00	1.00
PREVENTABLE	0.215	0.00	0.412	0.00	1.00
FATALITY	0.453	0.00	0.499	0.00	1.00
HISTORY	0.251	0.00	0.435	0.00	1.00
REPUTATION	6.290	6.290	0.949	3.090	9.140
PUBLICITY [0, +2]	0.469	0.492	3.398	−6.000	19.657
PUBLICITY [0, +5]	0.611	0.370	3.112	−6.000	19.560
PUBLICITY [0, +10]	0.676	0.319	2.943	−6.000	19.414
PUBLICITY [0, +15]	0.636	0.248	2.937	−6.000	19.352
PUBLICITY [0, +20]	0.605	0.228	2.920	−6.000	19.227
VOLUME [0, +2]	0.758	0.375	1.971	−2.761	8.512
VOLUME [0, +5]	1.343	0.618	3.673	−5.054	17.716
VOLUME [0, +10]	1.927	0.885	6.016	−11.858	33.099
VOLUME [0, +15]	2.535	1.162	8.100	−15.518	45.378
VOLUME [0, +20]	2.896	1.550	10.254	−20.965	56.257
SIZE	9.918	9.922	0.778	7.628	11.606
LEVERAGE	0.672	0.662	0.282	0.112	3.950
MARKET-TO-BOOK	4.694	1.837	11.392	0.003	96.140

Table 5 Correlations

	1	2	3	4	5	6	7	8	9	10	11
1. CAR	–										
2. APOLOGY	–0.26	–									
3. PREVENTABLE	–0.14	0.19	–								
4. FATALITY	–0.13	–0.29	–0.04	–							
5. HISTORY	0.12	0.02	0.14	0.08	–						
6. REPUTATION	0.09	–0.09	–0.06	–0.08	–0.16	–					
7. PUBLICITY	–0.38	0.10	0.16	0.18	–0.13	–0.09	–				
8. VOLUME	–0.53	0.31	–0.04	–0.05	–0.09	–0.11	0.43	–			
9. SIZE	0.14	–0.09	–0.07	–0.22	0.18	0.55	–0.22	–0.10	–		
10. LEVERAGE	0.10	0.13	0.18	–0.01	0.01	–0.19	–0.06	–0.08	–0.23	–	
11. MARKET-TO-BOOK	–0.16	–0.07	–0.04	–0.11	–0.06	0.03	–0.01	–0.18	0.09	–0.02	–

For variables measured at multiple event windows (CAR, APOLOGY, PUBLICITY, VOLUME), we report the largest correlation across the event windows

Bold correlations are significant at a 5% level

Event Study Results

We present the average CAR during the crisis events and the associated statistical tests in Table 6. The full sample is in Panel A along with results partitioned according to crisis type (preventable and non-preventable). In Panel B we focus on preventable crises partitioned by apology and non-apology. Similarly, in Panel C, the non-preventable crises are split into apology and non-apology subsets.

In the full sample, both the parametric and non-parametric tests show a significant stock price drop on average in each window ranging from -1.70% for the 2-day event window and steadily declining to -3.41% for the 15-day event window. The negative CAR becomes less pronounced, but still significant at the 20-day event window (-2.89%). This finding is consistent with prior literature (e.g., Knight and Pretty 1999). When the focus is on preventable crises in Panel A, the CARs are negative and significant for both parametric and non-parametric tests in all windows (except for the 20-day window sign test). We also observe a downward trend in CARs from the 2-day (-3.05%) until the 15-day window (-5.43%) and then an increase, but still negative (-4.74%) in the 20-day window. We observe a similar pattern for the non-preventable crisis subset: In each window, the CARs are negative for all the parametric and non-parametric tests except for the 2-day sign test. The CARs decline from two (-1.33%) to fifteen days (-2.86%), and the decline is less in the 20-day window (-2.38%). A comparison between the average CARs from the preventable and non-preventable cases shows that preventable crisis firms suffer larger losses in all event windows except the 20-day window for which the difference is not significant. Thus, in general, these observations are consistent with SCCT (Coombs and Holladay 1996, 2002; Coombs 2007)

and support Hypothesis 1: Firms' share prices are penalized more when their responsibility for the crisis increases.

In Panel B, the numbers indicate that apologizing for a preventable crisis increases the CARs significantly across the five event windows such that for most tests, the CARs are no longer significantly negative (albeit these results are based on 14 firms that apologize within 20 days of the crisis out of 48 preventable crises). Furthermore, not apologizing for a preventable crisis causes significant losses to share price. Initial losses in the first two days of 3.33% increase through the first 15 days (-7.45%) until a reversal by 20-days post-crisis (-6.88%). Again, the CARs are negative for both parametric and non-parametric tests in all windows (except for the 20-day window sign test). A difference-in-means test shows that for preventable crises, the average CAR is significantly higher for apologizing firms, which is consistent with the matching hypothesis of SCCT (Coombs and Holladay 1996; Coombs 2007) and EVT (Burgoon 1978, 1993; Kim 2014), and supports Hypothesis 2b.

Apologizing for a non-preventable crisis (Panel C), on the other hand, has a negative effect on CAR in all windows except for the 2-day sign test, which provides evidence that mismatching the response type to the crisis responsibility is detrimental to the share price for non-preventable crises, supporting Hypothesis 2a. Unlike the CARs in Panels A and B, losses are worst for the 2-day window (-13.18%), and gradually increase to the 20-day window (-7.06%). Although still significantly negative (except for the sign test), the CARs for the non-apology response to a non-preventable crisis are significantly higher than the apology response group. Here we see a small negative shock in the 2-day window (-0.76%), which follows the pattern of declining through the 15-day window (-2.23%), and slightly reversing by the 20-day window (-1.64%).

Table 6 Event study results: cumulative abnormal returns (day 0 = crisis date)

WINDOW	FULL SAMPLE				PREVENTABLE CRISIS				NON-PREVENTABLE CRISIS				DIFFERENCE-IN-MEANS	
	N	CAR (%)	CSECT TEST	SIGN TEST	N	CAR (%)	CSECT TEST	SIGN TEST	N	CAR (%)	CSECT TEST	SIGN TEST	CAR (%)	t test
PANEL A: partitioned by crisis type														
[0, +2]	223	-1.70	-4.15 (<0.01)	2.93 (<0.01)	48	-3.05	-4.72 (<0.01)	-3.44 (<0.01)	175	-1.33	-2.72 (0.01)	-1.50 (0.13)	-1.72	-2.13 (0.04)
[0, +5]	223	-2.66	-4.91 (<0.01)	-4.00 (<0.01)	48	-4.75	-4.45 (<0.01)	-4.31 (<0.01)	175	-2.09	-3.37 (<0.01)	-2.41 (0.02)	-2.66	-2.15 (0.03)
[0, +10]	223	-2.83	-4.61 (<0.01)	-3.60 (<0.01)	48	-5.10	-4.16 (<0.01)	-3.44 (<0.01)	175	-2.21	-3.15 (<0.01)	-2.26 (0.02)	-2.89	-2.05 (0.04)
[0, +15]	223	-3.41	-4.83 (<0.01)	-3.46 (<0.01)	48	-5.43	-3.28 (<0.01)	-2.87 (<0.01)	175	-2.86	-3.69 (<0.01)	-2.41 (0.02)	-2.57	-1.40 (0.17)
[0, +20]	223	-2.89	-3.63 (0.03)	-2.39 (0.02)	48	-4.74	-2.57 (0.01)	-1.13 (0.26)	175	-2.38	-2.71 (0.01)	-2.11 (0.04)	-2.36	-1.16 (0.25)
PANEL B: preventable crises partitioned by apology/non-apology														
PANEL C: non-preventable crises partitioned by apology/non-apology														
PANEL D: preventable crises partitioned by apology/non-apology														
[0, +2]	8	-1.64	-2.14 (0.03)	-1.31 (0.19)	40	-3.33	-4.41 (<0.01)	-3.19 (<0.01)	169	1.69	1.57 (0.13)	1.80 (0.09)	2.93 (0.01)	2.25 (0.03)
[0, +5]	10	-1.75	-1.04 (0.30)	-1.78 (0.08)	38	-5.54	-4.43 (<0.01)	-3.93 (<0.01)	3.79	6.15	6.92	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)
[0, +10]	13	-0.62	-0.30 (0.77)	-1.23 (0.22)	35	-6.77	-4.79 (<0.01)	-3.29 (<0.01)	6.15	6.92	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)
[0, +15]	14	-0.53	-0.23 (0.82)	-1.45 (0.15)	34	-7.45	-3.61 (<0.01)	-2.47 (0.01)	6.92	7.33	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)
[0, +20]	14	0.45	0.14 (0.89)	0.15 (0.88)	34	-6.88	-3.21 (<0.01)	-1.44 (0.15)	7.33	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)	1.87 (0.07)
PANEL E: non-preventable crises partitioned by apology/non-apology														
[0, +2]	8	-13.18	-2.14 (0.03)	-1.32 (0.19)	167	-0.76	-2.02 (0.04)	1.25 (0.21)	-12.42	-2.01 (0.08)	-2.21 (0.05)	-2.14 (0.05)	-1.78 (0.09)	-1.73 (0.10)
[0, +5]	14	-11.71	-2.48 (0.01)	-2.57 (0.01)	161	-1.25	-2.54 (0.01)	-1.75 (0.08)	-10.46	-6.92	-5.03	-5.42	-5.42	-5.42
[0, +10]	21	-8.30	-2.70 (0.01)	-2.74 (0.01)	154	-1.38	-2.28 (0.02)	-1.39 (0.16)	-6.92	-5.03	-5.42	-5.42	-5.42	-5.42
[0, +15]	22	-7.26	-2.67 (0.01)	-3.33 (<0.01)	153	-2.23	-2.83 (0.01)	-1.31 (0.19)	-5.03	-5.42	-5.42	-5.42	-5.42	-5.42
[0, +20]	24	-7.06	-2.34 (0.02)	-2.35 (0.02)	151	-1.64	-1.84 (0.07)	-1.33 (0.18)	-5.42	-5.42	-5.42	-5.42	-5.42	-5.42

Coefficients are reported first followed by the *p* value in brackets

Bold numbers are significant at 5%

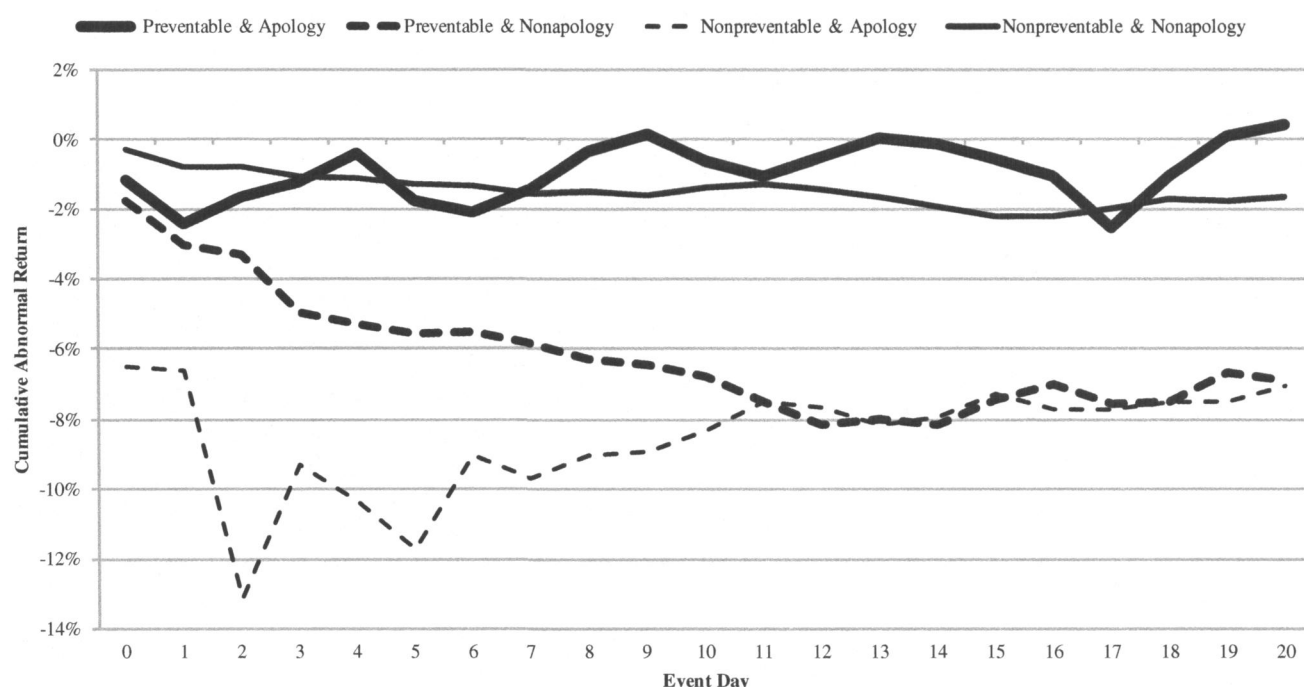


Fig. 1 Cumulative abnormal return for apology and non-apology response to preventable and non-preventable crises (day 0=crisis date)

Figure 1 provides a visual representation of the results in Table 6. In Fig. 1, thick lines are used to represent the preventable crises while thin lines are used to represent the non-preventable crises. Additionally, solid lines are used to represent crisis response strategies that are matched to the crisis type while the dotted lines represent crisis response strategies that are mismatched to the crisis type. The chart clearly illustrates that (not) apologizing when the firm is (not) responsible leads to lower losses than the opposite strategy. This figure, along with the results in Table 6, provides preliminary evidence that the response strategy should match the type of crisis as is recommended by SCCT (Coombs 2007) so that individuals' expectations for how the organization should respond are met (Burgoon 1993). To mitigate shareholder losses, firms should apologize for preventable crises (for which the firm is responsible), but not for non-preventable crises, and vice versa.

The event study supports the notion that as a firm's responsibility increases, the market reacts more negatively, resulting in a decrease in share price. We find evidence that the nature of the company's response to the crisis can have a substantial effect on share price. Firms that match their response strategy to their degree of responsibility for the crisis, as outlined by the SCCT, experience smaller losses compared to those that mismatch their responses. Based on these results, we support Hypotheses 1, 2a, and 2b.

Cross-Sectional Regression Results

While the results of Table 6 suggest that, at least on average, the non-apologizing firms would have been better off had they apologized, other costs and firm characteristics could have played a role in their decision. We turn to cross-sectional regression analysis to further test Hypotheses 1, 2a, and 2b while controlling for other crisis and firm characteristics that could affect CAR. Table 7 reports the results of estimating the crisis event sample for each event window. Consistent with Hypothesis 1, we find a significant negative association between PREVENTABLE and CAR across the five windows. The negative impact is weakest in the 2-day event window (-0.02) and deepens until the 10-day window (-0.05). Considering that the average CARs for the crisis event sample range from -0.0170 to -0.0341 (see Table 4), the direct effects from firm responsibility (PREVENTABLE) are also economically significant, *ceteris paribus*.

Apologizing (APOLOGY) has an adverse direct effect on CAR across all five windows with coefficients ranging from -0.04 to -0.10 and significant results for the first three windows. However, both SCCT and our preliminary results reported in Table 6 clearly show that the context of firm responsibility is critical for assessing the effect of apologizing. As such, we also interact apology with responsibility (APOLOGY*PREVENTABLE). Thus the coefficient on APOLOGY measures the direct effect from apologizing for non-preventable crises

Table 7 Cross-sectional regression results

Parameter ^a	Predicted sign	Cumulative abnormal return									
		[0, +2]	Sig.	[0, +5]	Sig.	[0, +10]	Sig.	[0, +15]	Sig.	[0, +20]	Sig.
INTERCEPT		0.01		0.06		−0.04		0.05		0.17	
		<i>0.06</i>		<i>0.10</i>		<i>0.11</i>		<i>0.11</i>		<i>0.13</i>	
PREVENTABLE (B1)	−	−0.02	***	−0.04	***	−0.05	***	−0.04	**	−0.05	**
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>	
APOLOGY (B2)	−	−0.10	**	−0.08	**	−0.05	*	−0.04		−0.04	
		<i>0.05</i>		<i>0.03</i>		<i>0.03</i>		<i>0.02</i>		<i>0.03</i>	
APOLOGY * PREVENTABLE (B3)	+	0.11	**	0.10	**	0.10	**	0.08	**	0.10	**
		<i>0.05</i>		<i>0.05</i>		<i>0.04</i>		<i>0.04</i>		<i>0.05</i>	
FATALITY	−	−0.01		−0.03	**	−0.03	**	−0.03	**	−0.03	*
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>	
HISTORY	−	0.01		0.01		−0.01		−0.01		0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>	
REPUTATION	+	−0.01		−0.01		0.01		−0.01		0.01	
		<i>0.00</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>	
PUBLICITY	−	−0.02		−0.03	*	−0.04	**	−0.04	***	−0.06	***
		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.02</i>		<i>0.02</i>	
VOLUME	−	−0.01	***	−0.01	***	−0.01	***	−0.01	**	−0.01	**
		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>	
SIZE	+	0.01		0.01		0.01		0.01		−0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>	
LEVERAGE	−	0.01		−0.01		0.04	**	0.04	*	0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.03</i>	
MARKET-TO-BOOK	?	−0.01	***	−0.01	***	−0.01	***	−0.01	***	−0.01	**
		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>	
Industry and year effects		Yes		Yes		Yes		Yes		Yes	
N		223		223		223		223		223	
# of apologetic firms		16		24		34		36		38	
R-squared		42.55%		41.54%		30.34%		25.51%		23.81%	
Wald test of change in R-squared for different models ^b											
Controls only		34.45%		36.14%		25.97%		23.24%		21.26%	
Controls + B1 & B2		38.31%	**	38.55%	**	27.78%	*	24.10%		21.92%	
Controls + B1 & B2 & B3		42.55%	**	41.54%	**	30.34%	**	25.51%	**	23.81%	**

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. P values reported at the top of the table are 2-sided while p values at the bottom of the table for the Wald test are one-sided

^aThe number reported on the same line as the parameter name is the coefficient and the number reported below in italics is the robust standard error

^bWe report R^2 for a base model (controls only), a model that includes main effects of our variables of interest plus controls, and finally the full model which adds the interaction term. Significance reported is based on a Wald test of the change in R^2 , adjusted for degrees of freedom, by adding additional variables

and its negative values are consistent with the results in Table 6 and further support Hypothesis 2a. The interaction term (APOLOGY*PREVENTABLE) has a positive and significant effect on CAR, suggesting that apologizing for preventable crises tends to mitigate the negative effect of the crisis, which supports Hypothesis 2b. To further investigate the roles played by our variables of interest, we run a pair of Wald tests for each event window to test the impact

of our focus variables on model fit compared to a baseline model that includes only control variables (see the second part of Table 7). The addition of APOLOGY and PREVENTABLE to the controls-only model leads to a relatively weak increase in the R-squared (significant at 5% for the 2- and 5-day windows and at 10% for the 10-day window). However, when we add the interaction term APOLOGY * PREVENTABLE, the R^2 increases significantly (at 5%) for

all event windows, indicating that the value of apologizing depends crucially on the firm's fault regarding the occurrence of the crisis.

Looking at the firm- and crisis-level control variables in our regression, we find that deaths caused by or related to the crisis (FATALITY) is significantly negatively associated with CAR across four of the five windows, suggesting that share price tends to decrease further for more severe crises. These results are consistent with prior literature, which finds that fatalities increase the loss of share value for a firm (Walker et al. 2005). We find no significant effect from the company having a previous history of crises (HISTORY) or from the company's previous reputation (REPUTATION).¹⁷ We do observe that CAR is negatively related to abnormal publicity (PUBLICITY). As expected, abnormal publicity reduces share price significantly starting in the 5-day window. CAR's negative association with publicity gets stronger (more negative) from the 10-day (-0.04) to the 20-day window (-0.06), so even though the abnormal publicity is declining between the 10- to 20-day windows, its effect on the abnormal return is not. Also consistent with our expectations, CAR is negatively related to abnormal trading volume (VOLUME) across all five event windows. As the crisis brings more attention to the firm, trading volume goes up and share price falls. The greatest impact is in the 2-day window (-0.01), steadily declining to the 20-window (< -0.01). CAR is not significantly affected by SIZE, and the effect of LEVERAGE is also insignificant except in the 10- and 15-day windows, where it has a positive effect. This observation seems counter intuitive as more leverage means greater financial risk, and perhaps a more difficult time surviving the crisis, although it is possible that we may be observing an asset substitution effect where, as the firm becomes riskier there is a transfer of wealth from debt to equity holders. Finally, CAR is negatively related to MARKET-TO-BOOK in all windows, consistent with the idea that growth firms may have more to lose from a crisis.

The correlations among some of the control variables are statistically and economically significant. For example, SIZE and REPUTATION have a correlation of 0.55. We use variance inflation factors (VIF) to gauge whether multicollinearity could be affecting any of the estimated coefficients. A VIF of 10 signals potential problems; however, our largest VIF is 2.76, so multicollinearity is not an issue. We likewise checked to see if our results were driven by outliers in the data. We use Cook's Distance (D)

to identify cases that could be exhibiting undue influence on our results. Using a threshold score of $4/n$ (0.179), we find that on average 18 observations in our sample may have undue influence on our results. For robustness, we rerun all our analyses dropping those observations with D scores higher than the threshold. All findings reported above were still significant, except in the 20-day window regression where APOLOGY * PREVENTABLE becomes insignificant.

To summarize, the cross-sectional regression results for the crisis sample illustrate that after controlling for crisis and firm-level potential influences, Hypotheses 1, 2a, and 2b, are maintained. In particular, we continue to find support for the SCCT notion of matching crisis response to crisis responsibility, although we acknowledge that these conclusions are based on a relatively small sample of 38 firms that apologize within 20 days of the crisis.

Endogeneity

The SCCT maintains that the causal relation between a crisis occurring and the firm choosing an appropriate response is unidirectional: A crisis happens, and then management decides what course of action to take in response to the crisis, with the share price responding to that decision. However, management's response strategy may be influenced by the market's response to the crisis, thereby creating reverse causality with CAR affecting APOLOGY. While including fixed effects in the regression models helps mitigate this concern, to address the possibility of endogeneity further, we estimate a 2-stage least squares (2SLS) regression using a dummy variable that indicates whether the stock trades on the NYSE as a potential instrumental variable. The NYSE is considered a more prestigious exchange than either Nasdaq or Amex (Kedia and Panchapagesan 2011) and managers cite increased visibility as a rationale for a preference for NYSE membership (Baker et al. 1999). Consequently, an NYSE listed firm may feel secure in the stature granted by its NYSE membership and thus less concerned with apologizing. Since APOLOGY is a binary variable, we follow the steps outlined by Wooldridge (2010, Ch. 20) to conduct 2SLS. Wooldridge shows that this procedure maintains valid standard errors. We begin by predicting the likelihood of APOLOGY using a probit regression on the instrument plus all control variables in Eq. (1).¹⁸ We follow Semadeni et al. (2014) in evaluating this potential instrument by examining first stage as well

¹⁷ If the reputation of missing values is replaced by 0 rather than the industry average reputation, then reputation does have a positive effect on CAR, but does not affect our inferences regarding our hypotheses.

¹⁸ We exclude industry and year fixed effects in the first stage because APOLOGY does not have sufficient variation among different time periods and industries, which reduces the sample size by more than half for most event windows.

Table 8 Two-stage least squares regression (2nd stage results)

Parameter ^a	Predicted Sign	Cumulative abnormal return									
		[0, +2]	Sig	[0, +5]	Sig	[0, +10]	Sig	[0, +15]	Sig	[0, +20]	Sig
INTERCEPT		0.01		−0.02		−0.03		−0.03		0.06	
		<i>0.05</i>		<i>0.08</i>		<i>0.09</i>		<i>0.10</i>		<i>0.12</i>	
PREVENTABLE (B1)	−	−0.02	**	−0.05	***	−0.08	***	−0.08	***	−0.08	***
		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.03</i>		<i>0.03</i>	
Pr{APOLOGY} (B2)	−	−0.07		0.01		−0.06		0.01		0.11	
		<i>0.06</i>		<i>0.04</i>		<i>0.07</i>		<i>0.09</i>		<i>0.10</i>	
Pr{APOLOGY} * PREVENTABLE (B3)	+	0.11		0.14	**	0.23	***	0.17	*	0.13	
		<i>0.07</i>		<i>0.07</i>		<i>0.09</i>		<i>0.09</i>		<i>0.10</i>	
FATALITY	+	−0.01		−0.01		−0.01		−0.01		0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.02</i>	
HISTORY	−	0.01		0.01		−0.01		−0.01		0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>	
REPUTATION	+	−0.01	*	−0.01	*	0.01		−0.01		−0.01	
		<i>0.00</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>	
PUBLICITY	−	−0.02	**	−0.03	**	−0.04	**	−0.04	**	−0.06	***
		<i>0.01</i>		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.02</i>	
VOLUME	−	−0.01	***	−0.01	***	−0.01	***	−0.01	**	−0.01	***
		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>	
SIZE	+	0.01		0.01		0.01		0.01		−0.01	
		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>		<i>0.01</i>	
LEVERAGE	−	0.01		−0.02		0.01		0.02		0.01	
		<i>0.01</i>		<i>0.02</i>		<i>0.02</i>		<i>0.03</i>		<i>0.03</i>	
MARKET-TO-BOOK	?	−0.01		−0.01	***	−0.01	**	−0.01	***	−0.01	**
		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>		<i>0.00</i>	
Industry & year effects		No		No		No		No		No	
N		223		223		223		223		223	
# of apologetic firms		16		24		34		36		38	
R-squared		37.82%		26.09%		18.58%		10.64%		<i>Not reported</i>	
Durbin (<i>p</i> -value)		0.80		0.02		0.17		0.15		0.05	
Wu-Hausman (<i>p</i> -value)		0.81		0.02		0.19		0.16		0.06	

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, two-tail tests of significance

^aThe number reported on the same line as the parameter name is the coefficient and the number reported below in italics is the robust standard error

as Hausman Wu statistics.¹⁹ In the first stage probit regressions, the Wald Chi-Square test on the exchange dummy is significant at < 0.005 in all five windows, indicating that this variable has a significant impact on APOLOGY. In addition, following Kennedy (2008) we calculate the correlations between the instrument and the endogenous variable (APOLOGY) and the correlations between the instrument and the OLS error from Eq. (1). A good instrument should have a high correlation with the endogenous

variable and no correlation with the residual. Our correlations between the exchange dummy and APOLOGY range between -0.18 and -0.31 , depending on the window. Semadeni et al. (2014) classify the instrumental variable as moderate if the correlation is 0.33 and weak if the correlation is 0.10. The correlations of the exchange dummy with the OLS residual range from -0.03 to $+0.07$ and are never significant. These findings support proceeding with the use of the exchange dummy as the instrumental variable for APOLOGY. We also construct an instrument for the interaction by taking the estimated APOLOGY variable and multiplying it by the PREVENTABLE dummy variable. The 2SLS regressions replace APOLOGY and APOLOGY*PREVENTABLE by the instruments

¹⁹ Semadeni et al. (2014) use an F test and ordinary regression analysis to determine if instruments are weak in the first stage regression. Our endogeneity issue surrounds a binary variable (APOLOGY) and thus we used probit analysis. We have adapted our tests appropriately.

$\Pr\{\text{APOLOGY}\}$ and $\Pr\{\text{APOLOGY}\} * \text{PREVENTABLE}$. This procedure is repeated for each of the five CAR event windows and the results from this second stage are reported in Table 8.²⁰ The second stage results generally confirm our main conclusions from using OLS regression. Preventable crises have a significantly lower CAR (coefficients range from -0.02 to -0.08), and firms that apologize for preventable crises have a significantly higher CAR as measured by the $\Pr\{\text{APOLOGY}\} * \text{PREVENTABLE}$ interaction (coefficients range from 0.11 to 0.23). However, the coefficients for $\Pr\{\text{APOLOGY}\}$ are not significant for any of the event windows, suggesting that apologizing for non-preventable crises is not driving CAR for these firms. Furthermore, the coefficients for the FATALITY dummy variable are also not significant, so more serious crises do not seem to have lower CARs here. The results for the other control variables are similar to the OLS cases.

Additional Analysis: Apology Event Windows

As an alternative test of the moderating effect of an apology, we perform an additional event study using only apology firms. We use the same procedure as described in the method section with the notable exception of using the apology date as the event date instead of the crisis, although the estimation window is still calculated prior to the crisis to avoid any confounding effects in the period between the crisis date and the apology date. Our sample for this test consists of the same 38 apologetic firms used in our analysis above with the addition of 8 firms that apologized *at least* 20 days *after* the onset of the crisis but no longer than 60 days after the crisis. For robustness, we ran an event study that excluded these additional 8 firms and find that our inferences are unchanged. Given the small size of the sample, we do not perform any cross-sectional analysis in this case. Similar to the prior event study, we divide the sample into subsets based on preventable and non-preventable crises to allow us to compare the effects of an apology being a match or mismatch to the degree of responsibility.

We present the results of the apology event study in Table 9. Looking at the full sample of apology firms we again see the pattern of an initial negative shock in share price that begins to reverse by the tenth day. When the sample is portioned into preventable and non-preventable crises, two different trends appear (see Fig. 2). For the preventable crises, for which the responsibility and response are matched

Table 9 Event study results: cumulative abnormal returns (day 0 = apology date)

Window	Full sample				Preventable				Non-preventable				Difference-in-means	
	N	CAR (%)	CSECT test	SIGN test	N	CAR (%)	CSECT test	SIGN test	N	CAR (%)	CSECT test	SIGN test	CAR (%)	t test
[0,+2]	46	-2.42	-1.92 (0.06)	-0.68 (0.50)	17	-1.43	-1.54 (0.12)	-1.57 (0.12)	29	-3.00	-1.55 (0.12)	0.35 (0.73)	1.57	0.73 (0.47)
[0,+5]	46	-3.33	-2.08 (0.04)	-1.27 (0.21)	17	-0.09	-0.08 (0.93)	-0.11 (0.91)	29	-5.23	-2.17 (0.03)	-1.51 (0.13)	5.14	1.96 (0.06)
[0,+10]	46	-3.10	-1.76 (0.08)	-1.27 (0.21)	17	0.54	0.24 (0.81)	0.38 (0.71)	29	-5.24	-2.18 (0.03)	-1.88 (0.06)	5.78	1.75 (0.09)
[0,+15]	46	-3.04	-1.67 (0.09)	-0.97 (0.33)	17	0.08	0.04 (0.97)	-0.11 (0.91)	29	-4.86	-1.87 (0.06)	-1.13 (0.26)	4.94	1.51 (0.14)
[0,+20]	46	-1.57	-0.76 (0.45)	0.50 (0.61)	17	3.46	1.51 (0.13)	2.32 (0.02)	29	-4.51	-1.58 (0.11)	-1.13 (0.26)	7.98	2.18 (0.03)

Coefficients are reported first, then the *p*-value in brackets
Bold numbers are significant at 5%

²⁰ We perform a Durbin test and a Wu-Hausman test for endogeneity on each of the five event windows and find that our instrument is valid for the 5-day and 20-day event windows. For the other event windows, the standard OLS regression in Eq. (1) performs just as well or better than the 2SLS regression.

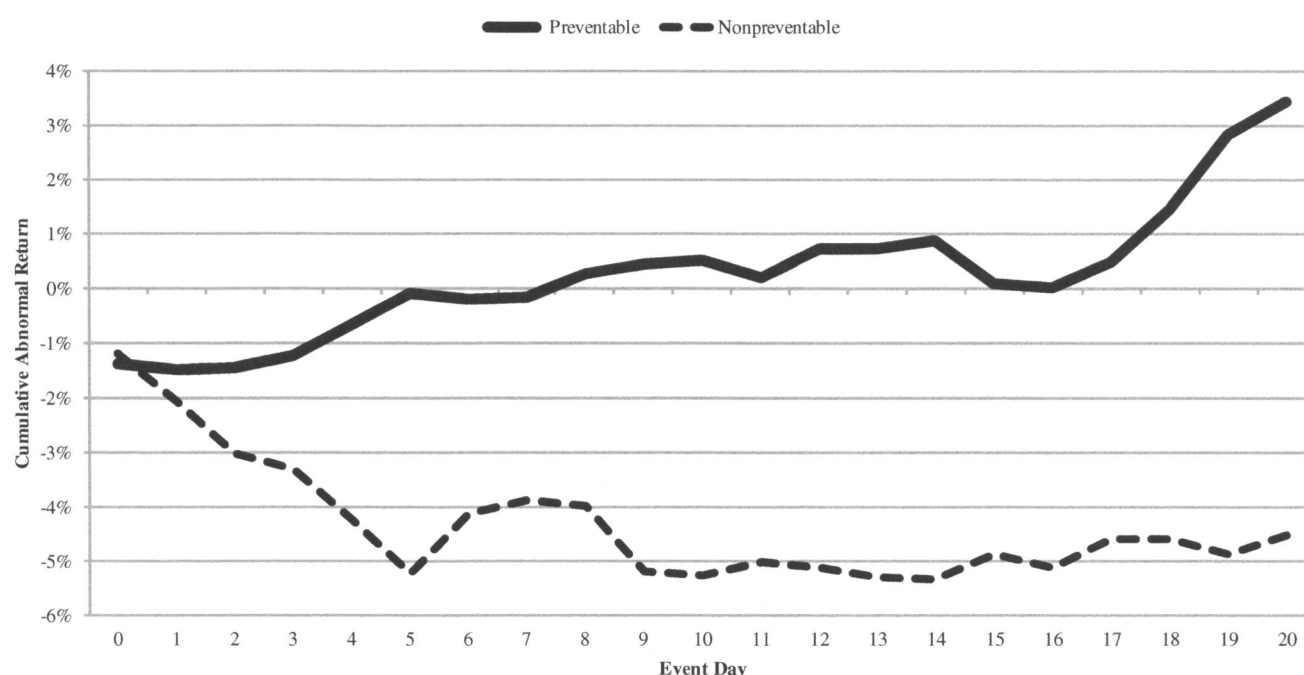


Fig. 2 Cumulative abnormal return for apologetic preventable and non-preventable crises (day 0 = apology date)

in this sample of apologizing firms, share price is initially pushed downward (2-day = -1.43%) but by day 5 it recovers to close to 0%, where it remains until the 20-day window, when it is significantly greater than zero on average (3.46%). For the non-preventable crises, for which the responsibility and response are mismatched, the price drops initially (2-day = -3.00%) and remains low, without showing similar signs of recovery (20-day = -4.51%).

The results from the second event study corroborate our initial findings. Apologizing when it is not warranted according to SCCT leads to a strongly negative reaction by investors, whereas apologizing when the firm is responsible leads to a neutral and eventually positive response by investors. These findings further support Hypotheses 2a and 2b.

Discussion

Our comprehensive sample of the 223 firms with non-financial organizational crises between 1983 and 2013, for which shareholders are not the direct victims of the crisis, provides evidence that the nature of the firm's response can influence its stock price. Faced with a company crisis, shareholders make a judgment about the firm's level of responsibility based on the type of crisis, the severity, the history of crises for the company, and the prior relationship reputation (Coombs 2007). Once the shareholder makes this judgment, she/he forms an expectation of the appropriate firm reaction. When the firm acts, it will either meet or fail to meet the

investors' expectations based on matching or mismatching the crisis response to the company's degree of responsibility. Expectancy Violations Theory (Burgooon 1978, 1993) posits that when an individual's expectations are not met, it leads to an affective reaction (anger), which influences their subsequent behavior. Consistent with prior crisis research using this theory (Kim 2014), we find that investors react negatively when the firm's actions are contrary to their expectations.

Drawing on the EVT we explain the significant difference in CAR between the apology (match) and non-apology (mismatch) groups, Table 6 Panel B, as shareholders reacting to the violation of their expectations of firm appropriate behavior in light of a preventable crisis. In our results, the loss of shareholder value is not necessarily due to the shareholders' affective response to the lack of apology, since shareholders are not the victims of the crisis, but rather the market's assessment of the costs (or net costs) resulting from disaffected stakeholders. In other words, the affected stakeholders' expectations are violated, resulting in additional costs of doing business, which are recognized by the shareholders and market participants. Similarly, in Table 6 Panel C, we explain the significant difference in CAR between the apology (mismatch) and non-apology (match) groups as shareholders reacting to the violation of their expectation of the firm's appropriate acknowledgement of their role in the non-preventable crisis. In this case, the stakeholders are not disaffected by the apology per se, but rather the act of apologizing may cause them (and other stakeholders) to attribute a

greater culpability to the firm than is warranted. The results are visually supported by the clustering of matched and mismatched crisis responses in Fig. 1. We use cross-sectional regression analysis to control for other factors that could affect how the market reacts to a firm's crisis. The results in Tables 7 and 8 generally support the findings in Table 6. Supplemental analysis in Table 9 and presented visually in Fig. 2 corroborates these findings, showing that apologies are helpful (harmful) to shareholders when offered in response to a preventable (non-preventable) crisis. Despite these consistent results from various methods, we reiterate that our conclusions are based on a relatively small sample of firms, and we recognize that our results should be interpreted with caution.

Synthesizing these results suggests that, in our sample, from the firm value perspective, the appropriate response to a crisis depends critically on the degree of company responsibility for the crisis and matching the firm reaction to the level of responsibility. Failing to apologize when the crisis is preventable exacerbates the situation but apologizing when it is unnecessary is also harmful. These market response results differ from prior experimental literature, which finds that failing to match crisis response to crisis type has no significant impact on the individual's behavior (Claeys et al. 2010). This difference can be explained by EVT, which suggests that mismatching should have a significant impact on share price.

Conclusion

We examine how stock price responds to non-financial organizational crises when shareholders are not the direct victims of the crisis, and how corporate apologies affect the price response. We use the Situational Crisis Communication Theory (SCCT) (Coombs 2007) and Expectancy Violation Theory (EVT) (Burgoon 1993) to motivate our hypotheses about the stock market reaction to crisis type and crisis response. Specifically, we hypothesize that market reaction should be more negative for preventable crises, and that apologizing should mitigate or exacerbate the negative market reaction depending on the appropriateness of the firm's response relative to its degree of involvement. We build a sample of 223 organizational crises over the period 1983–2013 based on crises used in prior research and hand-collected data from government agencies. We use an event study method accompanied by cross-sectional regression analysis to test our hypotheses.

Consistent with prior literature, our results show that investors react negatively to organizational crises. This negative reaction is worse when the organization is primarily responsible for causing the crisis and when the crisis involves fatalities (Marcus and Goodman 1991). If the

apology corresponds to a preventable crisis such as tampering with automobile odometers in the case of Chrysler, the market reacts positively to the apology, leading to a reduction in the negative share price response. On the other hand, apologizing for a non-preventable crisis, such as the hardware failure of an external satellite in the case of Blackberry, exacerbates the negative reaction.

Our outcomes for the interaction between apology and crisis responsibility provide support for the postulated matching of crisis response to firm responsibility within the SCCT (Coombs 2007). These findings are especially noteworthy in light of the controversy in the experimental methods literature as to the usefulness of matching crisis type and response. Early studies show that matching leads to positive results (Coombs and Holladay 1996), but more recent research finds no effect from matching or mismatching responses (Claeys et al. 2010). Using archival methods, we show that there is a matching effect (at least for shareholder who are not directly affected by the crisis): A firm apologizing for preventable and not apologizing for non-preventable crises helps mitigate the market's negative response to the crisis. While message involvement and message framing may affect the efficacy of crisis response matching (Claeys and Cauberghe 2014), we offer an alternative explanation using EVT. Stakeholders form expectations as to how the organization *should* respond to the crisis based on its degree of responsibility. If companies react in a way that violates stakeholder expectations, the negative shareholder response is exacerbated.

The findings of this paper are also useful to managers. The decision to apologize should be carefully vetted and considered in context since, as our results show, apologizing when it is not warranted could reduce shareholder wealth. However, managers should apologize when the firm is directly responsible for the crisis because failing to do so could also jeopardize investor wealth. The apology discussion should not be oversimplified nor should shareholder response be the only consideration. Future research should further investigate various nuances and explore a multifactor decision analysis that considers various stakeholders.

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Compliance with Ethical Standards

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